Computer Networks Lab

Cisco Packet Tracer — Assignment 5

All practicals must be completed inside **Cisco Packet Tracer**.
Each assignment requires **two screenshots** as submission.

Topic: Standard Access Control Lists (ACL)

How to Submit

- Complete the practical entirely inside **Cisco Packet Tracer**. No other tool is accepted.
- Save your file as `RollNo_Set4.pkt` (e.g. `20221234_Set4.pkt`) and submit it.
- Along with the file, submit exactly **two screenshots** as described at the end of the set.
- Screenshots must be clear and show all device labels and relevant details.

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Scenario

Fortuna Bank has three departments connected through a central router: **HR**, **Finance**, and **IT**. The IT department hosts a sensitive **Payroll Server** that stores salary records for every employee in the bank.

Until now, anyone on any department's network could reach the Payroll Server, and this has become a serious security concern. The bank's security officer, Mr. Devraj Rathore, has decided that **only the Finance department** should be allowed to access the Payroll Server — the HR department must be completely blocked from reaching it, while everything else on the network continues to work normally.

Since this rule only needs to filter traffic based on *where it came from*, Mr. Rathore asks you to implement the restriction using a **Standard Access Control List (ACL)**, placed as close to the destination (the Payroll Server's network) as possible.

Device	IP Address	Default Gateway
Router Gi0/0 (HR)	192.168.10.1	—
HR-PC1	192.168.10.10	192.168.10.1
HR-PC2	192.168.10.20	192.168.10.1
Router Gi0/1 (Finance)	192.168.20.1	—
Fin-PC1	192.168.20.10	192.168.20.1
Fin-PC2	192.168.20.20	192.168.20.1
Router Gi0/2 (IT/Server)	192.168.30.1	—
Payroll-Server	192.168.30.10	192.168.30.1
Subnet Mask: `255.255.255.0` on all devices		

Your Task

- Build the topology: one router with three interfaces, one switch per department (HR and Finance), the Payroll Server connected on the IT side, and two PCs each in HR and Finance.
- Assign all IP addresses and default gateways from the table, and activate every router interface.
- Before configuring any ACL, verify that **both** an HR PC and a Finance PC can successfully ping the Payroll Server.
- Configure a **Standard ACL** (number 1–99) on the router that:
  - **denies** traffic from the HR network (`192.168.10.0/24`), and
  - **permits** all other traffic.
- Apply this ACL on the correct interface and in the correct direction so that it filters traffic **only as it leaves towards the Payroll Server**, without blocking HR's access to anything else.

- Re-test connectivity: the HR PC's ping to the Payroll Server must now **fail**, while the Finance PC's ping to the Payroll Server must still **succeed**.

Submission

Submit the following **two screenshots**:

1. The completed **topology** in Packet Tracer Logical View with all devices connected and IP labels visible, along with the router's CLI showing the `show running-config` (or `show access-lists`) output confirming the standard ACL rule and the interface it is applied to.
2. A single screenshot showing **two Command Prompt windows** side by side — one from an HR PC showing the ping to the Payroll Server **failing** (Request timed out / Destination unreachable), and one from a Finance PC showing the ping to the same server **succeeding** with zero packet loss.

Note: A **Standard ACL** can only filter based on the *source IP address* — it has no idea about the destination. For this reason, Cisco's best practice is to place a standard ACL **as close to the destination network as possible** (here, on Gi0/2, outbound) so it does not accidentally block the HR network's access to other departments or the internet.